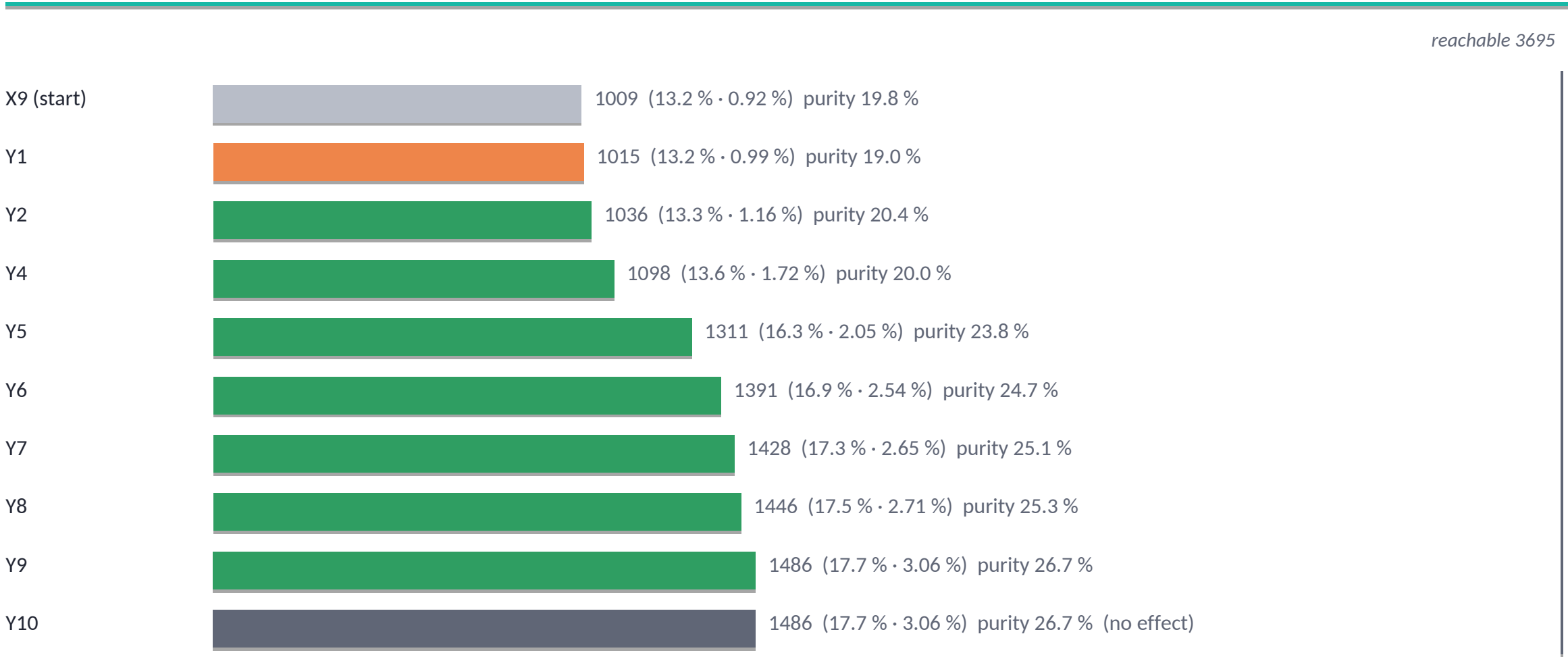
ZDC autoresearch, Phase 5: $\Lambda \to n\pi^0$ on MSF events

Find more $\Lambda \to n\pi^0$ in EICrecon on meson-structure events without a dirtier Λ sample

Started 2026-11-08 · one slide per try · rebuilt from the run's result files

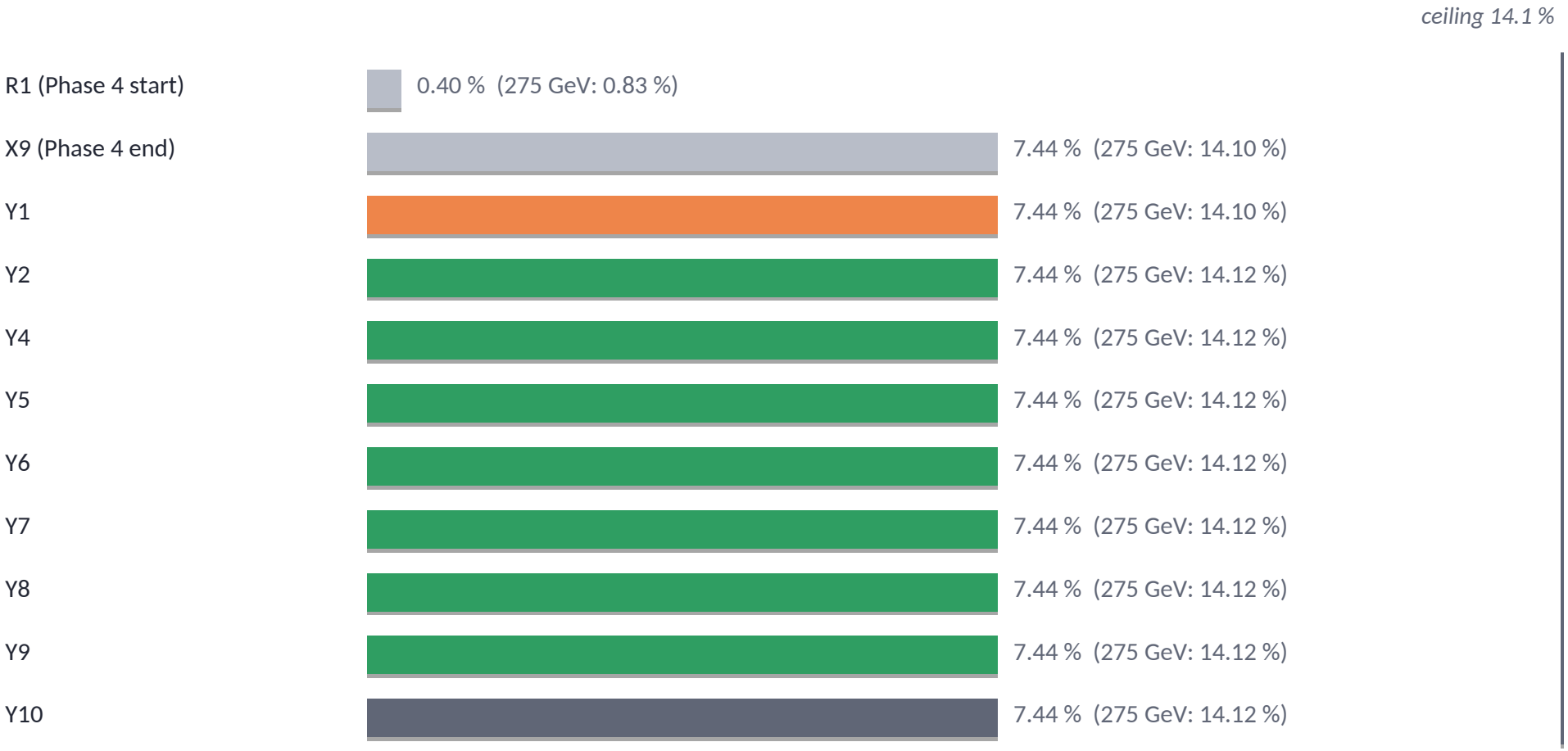
Score per try — Λ → nπ⁰ found on the MSF search halves, 9x275 + 9x130



Bar: found Λ pooled over the search halves (463). Text: found (yield 9x275 · 9x130), purity in the ±30 MeV window. Colour = verdict: green accepted, orange inconclusive, grey rejected. Rule: the reachable yield, pooled.

X9 1009 → Y9 1486 found (+47 %): 40 % of the reachable yield.

D4 gun score per try — S4a_tm per thrown Λ → nπ⁰, mean over 100–275 GeV



headline.csv (441); D4 reconstructed for every Phase 5 try at the close (D4P5_*, CP5_*). Reported, never selected on.

Phase 5 leaves the gun where Phase 4 left it: its gains are in the B0 regimes the gun barely reaches.

Phase 5 tries — Λ found on the search halves, step by step

try	change	9x275	9x130	score	purity	z	verdict
Y1	B0 ECal clusters split at local maxima	943 → 944	66 → 71	1009 → 1015	19.8 → 19.0 %	+1.2	INCONCLUSIVE
Y2	the Λ vertex with each daughter's lever arm	943 → 953	66 → 83	1009 → 1036	19.8 → 20.4 %	+4.2	ACCEPTED
Y4	B0 splitting retried on top of Y2	953 → 975	83 → 123	1036 → 1098	20.4 → 20.0 %	+7.0	ACCEPTED
Y5	the ZDC pairing with a photon in B0	975 → 1164	123 → 147	1098 → 1311	20.0 → 23.8 %	+13.3	ACCEPTED
Y6	sharper energy sharing between split B0 clusters	1164 → 1209	147 → 182	1311 → 1391	23.8 → 24.7 %	+8.1	ACCEPTED
Y7	the B0 photon energy law fitted on a B0 photon gun	1209 → 1238	182 → 190	1391 → 1428	24.7 → 25.1 %	+3.7	ACCEPTED
Y8	B0 split sharing narrowed again, 0.5 → 0.3 crystal	1238 → 1252	190 → 194	1428 → 1446	25.1 → 25.3 %	+3.4	ACCEPTED
Y9	the ZDC pairing with both photons in B0	1252 → 1267	194 → 219	1446 → 1486	25.3 → 26.7 %	+5.2	ACCEPTED
Y10	the Λ-mass term of the final choice, 112 → 40 MeV	1267 → 1267	219 → 219	1486 → 1486	26.7 → 26.7 %	+0.0	REJECTED (no effect)

Score 1009 → 1486 (+47 %), purity 19.8 → 26.7 %; each row is paired against the best of its time.

Phase 5 tries in Phase 4's columns — S4a_tm per thrown Λ → nπ⁰ on the D4 gun

S4a_tm per thrown Λ → nπ⁰	mean 100–275 GeV	at 275 GeV	MSF found, 9x275 / 9x130	verdict
R1 (PR 3064), start of Phase 4	0.404 %	0.832 %	—	
X9, end of Phase 4 = start of Phase 5	7.436 %	14.101 %	943 / 66	
Y1: B0 ECal clusters split at local maxima	7.436 %	14.101 %	944 / 71	INCONCLUSIVE
Y2: the Λ vertex with each daughter's lever arm	7.439 %	14.117 %	953 / 83	ACCEPTED
Y4: B0 splitting retried on top of Y2	7.439 %	14.117 %	975 / 123	ACCEPTED
Y5: the ZDC pairing with a photon in B0	7.439 %	14.117 %	1164 / 147	ACCEPTED
Y6: sharper energy sharing between split B0 clusters	7.439 %	14.117 %	1209 / 182	ACCEPTED
Y7: the B0 photon energy law fitted on a B0 photon gun	7.439 %	14.117 %	1238 / 190	ACCEPTED
Y8: B0 split sharing narrowed again, 0.5 → 0.3 crystal	7.439 %	14.117 %	1252 / 194	ACCEPTED
Y9: the ZDC pairing with both photons in B0	7.439 %	14.117 %	1267 / 219	ACCEPTED
Y10: the Λ-mass term of the final choice, 112 → 40 MeV	7.439 %	14.117 %	1267 / 219	REJECTED (no effect)
ceiling (perfect separation, P1b resolution)	14.1 %	27.8 %	reachable 43.6 % / 8.0 % of expected	

Phase 5 score — yield and purity per beam energy

```
N_expected = 0.358 × generated Λ   (one Λ per event)
N_found    = events whose true Λ → n pi0 is rebuilt:
             candidate within ±30 MeV of m_Λ, and
             n, γ, γ clusters ≥ 80 % from the true
             neutron and the two true photons
N_window   = events with any candidate in the window

yield  = N_found / N_expected
purity = N_found / N_window
```

X9 on D6c (40 000 events per beam)	9x275	9x130
N_found / N_expected	1840 / 14320	129 / 14320
yield	12.85 ± 0.28 %	0.90 ± 0.08 %
reachable (n in ZDC, both γ read)	43.6 %	8.0 %
N_found / N_window	1840 / 8749	129 / 1408
purity	21.03 ± 0.44 %	9.16 ± 0.77 %

Selector (463): keep a change when

- the score N_found(9x275) + N_found(9x130) on the search halves D6s rises: on identical Λ, b found only by the current best, c only by the change, $z = (c - b) / \sqrt{(b + c)} \geq 3$
- and the pooled purity does not fall by more than 2σ: $z_P = (P_{new} - P_{best}) / \sqrt{(\sigma_{new}^2 + \sigma_{best}^2)} \geq -2$
- D6h (files 41–80) is held out: every 5 tries the current best must reproduce its gain there at > 2σ
- fair play F1–F3 of Phase 4 (460) on the MSF outputs; a FAIL voids the try

Yield is pushed up; purity is a floor. A change that only cleans the sample is reported, not kept.

Where $\Lambda \rightarrow n\pi^0$ land and what X9 finds in each regime

regime (share of N_expected; X9 found in it)	9x275	9x130
all three in the ZDC	27.9 %; 1472 found (36.8 %)	1.3 %; 62 found (34.3 %)
n ZDC, γ B0 + γ ZDC	11.0 %; 366 found (23.3 %)	2.1 %; 66 found (22.0 %)
n ZDC, γ γ in B0	4.0 %; 1 found (0.2 %)	3.5 %; 1 found (0.2 %)
n ZDC, a γ in EEMC-P / LFHCAL	0.7 %; 0 found (0.0 %)	1.2 %; 0 found (0.0 %)
n ZDC, a γ lost	25.6 %; 0 found (0.0 %)	10.1 %; 0 found (0.0 %)
n in B0	6.6 %; 1 found (0.1 %)	30.0 %; 0 found (0.0 %)
n lost or elsewhere	1.1 %; 0 found (0.0 %)	23.3 %; 0 found (0.0 %)
Λ interacts in material / does not decay	17.2 % / 6.4 %	27.2 % / 0.2 %

At 9x275 the software can still win in the first four rows; at 9x130 the neutron misses the ZDC for 53 %.

Y1 — B0 ECal clusters split at local maxima

Change

- B0 island clustering keeps one cluster per island today (splitCluster false)
- with both photons in B0 (4.0 % of expected at 9x275) the pair is merged for 93 % of the Λ
- split at every hit above 30 MeV that exceeds its 8 neighbours
- shared hits weighted by exp(−d / 1 crystal) × peak energy (stock: η-φ distance, scale 1)
- three parameters on the X9 install; no code change

Prediction (written before launch)

- N_found on the search halves: 9x275 ≈ 920 → 960, 9x130 ≈ 65 → 95; purity flat or down < 1 point

search half	best	this try
9x275 N_found	943	944
9x275 purity	21.5 %	20.6 %
9x130 N_found	66	71
9x130 purity	9.4 %	9.5 %

paired: b 9, c 15, z +1.22; purity z_P -1.01

INCONCLUSIVE

fair play PASS

Y2 — the Λ vertex with each daughter's lever arm

Change

- the Λ line through the IP had slope $\Sigma p_i x_i / \Sigma p_i z_i$: right when n, γ, γ hit one depth
- a B0 photon ($z \approx 7$ m) beside ZDC daughters ($z \approx 35$ m) was weighted down by ~7/35
- now, for a vertex at depth z: $s_\perp = \Sigma w_i x_{i\perp} / (P + z \Sigma w_i)$, $w_i = p_i / (z_i - z)$, at every bisection step
- the depth search ends short of the nearest impact point (it ran on to the ZDC)
- 33 lines in LambdaReconstruction.cc; all-ZDC events keep the stock result

Prediction (written before launch)

- 9x275 920 → ~990, 9x130 ~65 → ~80; all-ZDC unchanged within ±10; purity up slightly

search half	best	this try
9x275 N_found	943	953
9x275 purity	21.5 %	21.7 %
9x130 N_found	66	83
9x130 purity	9.4 %	11.9 %

paired: b 7, c 34, z +4.22; purity z_P +0.71

ACCEPTED | **fair play PASS**

Y4 — B0 splitting retried on top of Y2

Change

- Y1 resolved both B0 photons for ~50 more Λ (9x275) and ~65 (9x130), but their mass failed
- Y2 made the γγ-in-B0 regime count: 0 → 5 found at 9x275, 1 → 15 at 9x130
- Y1's three B0 island parameters on the Y2 install

Prediction (written before launch)

- γγ in B0 found 5 → ~20 and 15 → ~35; N_found 953 → ~968, 83 → ~103; purity flat

search half	best	this try
9x275 N_found	953	975
9x275 purity	21.7 %	20.8 %
9x130 N_found	83	123
9x130 purity	11.9 %	15.3 %

paired: b 8, c 70, z +7.02; purity z_P -0.50

ACCEPTED **fair play PASS**

Y5 — the ZDC pairing with a photon in B0

Change

- the ZDC pairing assumed both photons in the ZDC: with one photon in B0 it still wrote two ZDC photons
- now each tagged ZDC cluster with each B0 photon candidate is a hypothesis too (Y2's vertex)
- a winning mixed hypothesis writes one ZDC photon; the rest of the ZDC is the neutron
- LambdaReconstruction pairs that photon with the B0 photon
- switch lambdaPairingMixed, off by default; off reproduces Y4 exactly

Prediction (written before launch)

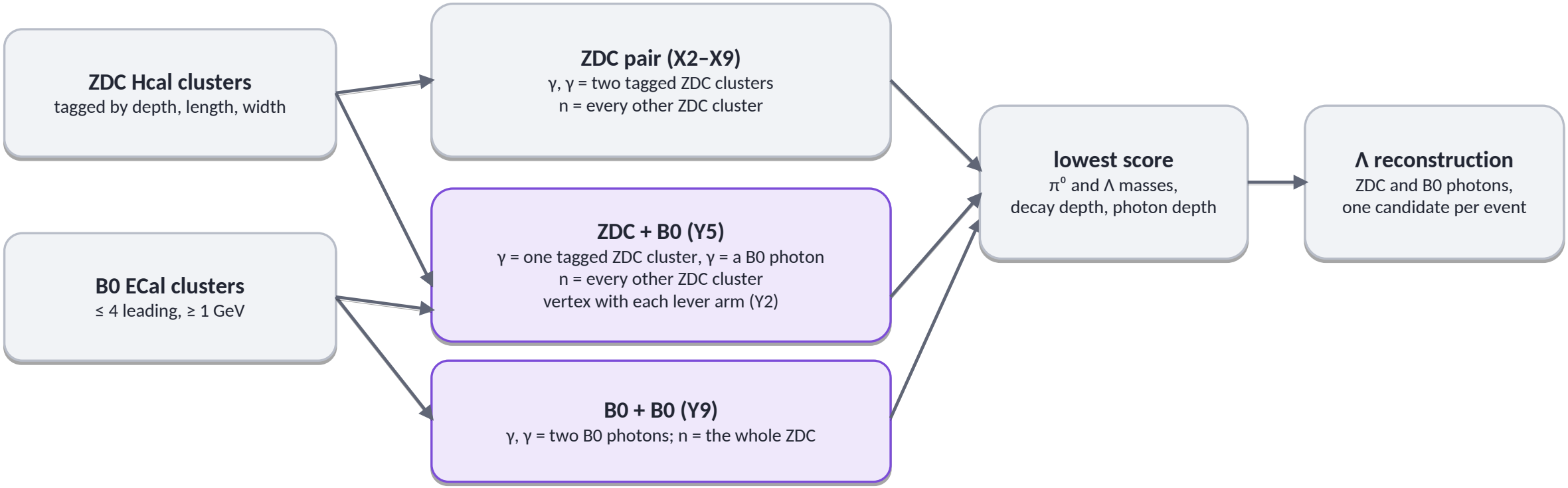
- emulated mixed regime 195 → 383 (9x275), 32 → 55 (9x130); N_found 975 → ~1150, 123 → ~145; purity up

search half	best	this try
9x275 N_found	975	1164
9x275 purity	20.8 %	24.6 %
9x130 N_found	123	147
9x130 purity	15.3 %	19.1 %

paired: b 22, c 235, z +13.29; purity z_P +4.84

ACCEPTED **fair play PASS**

Y5 and Y9 — photon hypotheses in the ZDC pairing



Why

- with one photon in B0 the old pairing still wrote two ZDC photons: the true one with a neutron fragment, or two fragments
- their energy left the neutron, and the true triplet could not form: another triplet chosen (186) or the ZDC photon missing (144) of 803
- emulation: Y5 383 instead of 195 in γ B0 + γ ZDC (474); Y9 +49 net with both photons in B0 (475)

Y6 — sharper energy sharing between split B0 clusters

Change

- with both photons in B0, Y5 still misses a photon for 198 of 264 Λ (9x275) and 151 of 264 (9x130)
- the photon's best B0 cluster is shared below purity 0.8 (~50 %) or merged with the other photon (~45 %)
- split hits are shared by exp(−d / s) × peak energy; s = 1.0 → 0.5 crystal
- one parameter on the Y5 install

Prediction (written before launch)

- γγ-in-B0 found 33 → ~50 (9x275), 57 → ~80 (9x130); N_found 1164 → ~1180, 147 → ~170; purity flat

search half	best	this try
9x275 N_found	1164	1209
9x275 purity	24.6 %	25.0 %
9x130 N_found	147	182
9x130 purity	19.1 %	23.5 %

paired: b 9, c 89, z +8.08; purity z_P +1.17

ACCEPTED | **fair play PASS**

Y7 — the B0 photon energy law fitted on a B0 photon gun

Change

- stock law 0.99·E^1.14 reads gun photons 11–48 % high at 5–40 GeV
- refit on D2b_photon_B0 (2–40 GeV, 7–20 mrad) with Y6's B0 clustering: 1.0888·E^1.0058
- every gun energy within 0.2 % of true; one parameter on Y6

Prediction (written before launch)

- γγ in B0 up, γ B0 + γ ZDC down (the stock overestimate cancelled the ZDC photon's 0.78); net small, sign uncertain

search half	best	this try
9x275 N_found	1209	1238
9x275 purity	25.0 %	25.2 %
9x130 N_found	182	190
9x130 purity	23.5 %	23.9 %

paired: b 31, c 68, z +3.72; purity z_P +0.38

ACCEPTED **fair play PASS**

Y8 — B0 split sharing narrowed again, 0.5 → 0.3 crystal

Change

- Y6 (1.0 → 0.5 crystal) gained +80, z +8.1
- both photons in B0 is still the regime with the most B0 losses: 87 and 97 of 264 found
- one parameter on Y7

Prediction (written before launch)

- N_found 1238 → ~1250, 190 → ~200; the gain may stop or turn (tail energy to the wrong maximum)

search half	best	this try
9x275 N_found	1238	1252
9x275 purity	25.2 %	25.4 %
9x130 N_found	190	194
9x130 purity	23.9 %	24.3 %

paired: b 5, c 23, z +3.40; purity z_P +0.28

ACCEPTED **fair play PASS**

Y9 — the ZDC pairing with both photons in B0

Change

- with both photons in B0 another triplet wins for 46 (9x275) and 53 (9x130) of 264 Λ
- Y5's pairing still writes ZDC fragments as photons there and takes their energy from the neutron
- third hypothesis: every pair of B0 photon candidates, the whole ZDC as the neutron
- if it wins, no ZDC photon is written; switch lambdaPairingExternalPair, off reproduces Y8

Prediction (written before launch)

- emulated net +21 / +28 in γγ-in-B0, −4 / −3 in γ B0 + γ ZDC; N_found 1252 → ~1268, 194 → ~219

search half	best	this try
9x275 N_found	1252	1267
9x275 purity	25.4 %	26.4 %
9x130 N_found	194	219
9x130 purity	24.3 %	28.7 %

paired: b 10, c 50, z +5.16; purity z_P +1.76

ACCEPTED **fair play PASS**

Y10 — the Λ-mass term of the final choice, 112 → 40 MeV

Change

- LambdaReconstruction ranks triplets across ZDC, B0 and EEMC-P by π^0 and Λ mass χ^2
- its Λ term had the width 0.1 m_Λ = 112 MeV; now 40 MeV, as the ZDC pairing
- after Y5 and Y9 another triplet wins for only 22 Λ on both search halves

Prediction (written before launch)

- N_found 1267 → ~1272, 219 → ~222; N_window up, purity down < 1 point

search half	best	this try
9x275 N_found	1267	1267
9x275 purity	26.4 %	26.4 %
9x130 N_found	219	219
9x130 purity	28.7 %	28.6 %

paired: b 0, c 0, z +0.00; purity z_P -0.02

REJECTED

fair play PASS

Checkpoint 1 — held-out MSF files and the D4 gun, X9 → Y6

N_found (yield)	X9, search	Y6, search	X9, held out	Y6, held out	reachable
9x275	943 (13.17 %)	1209 (16.89 %)	897 (12.53 %)	1151 (16.08 %)	43.6 %
9x130	66 (0.92 %)	182 (2.54 %)	63 (0.88 %)	165 (2.30 %)	8.0 %
purity, held out (pooled)			19.0 %	23.8 %	

held out: REPRODUCES

b 30, c 386, z +17.45; purity z_P +6.06

D4 Λ gun, S4a_tm per thrown Λ → nπ⁰ (reported, never selected on)

- X9 (end of Phase 4): 7.44 % mean, 14.10 % at 275 GeV; Y6: 7.44 % mean, 14.12 % at 275 GeV (ceilings 14.1 % / 27.8 %)

The search halves gained 38 %, the held-out halves 37 %: the accepted steps are no overfit.

Checkpoint 2 — held-out MSF files and the D4 gun, Y6 → Y9

N_found (yield)	Y6, search	Y9, search	Y6, held out	Y9, held out	reachable
9x275	1209 (16.89 %)	1267 (17.70 %)	1151 (16.08 %)	1206 (16.84 %)	43.6 %
9x130	182 (2.54 %)	219 (3.06 %)	165 (2.30 %)	205 (2.86 %)	8.0 %
purity, held out (pooled)			23.8 %	25.8 %	

held out: REPRODUCES

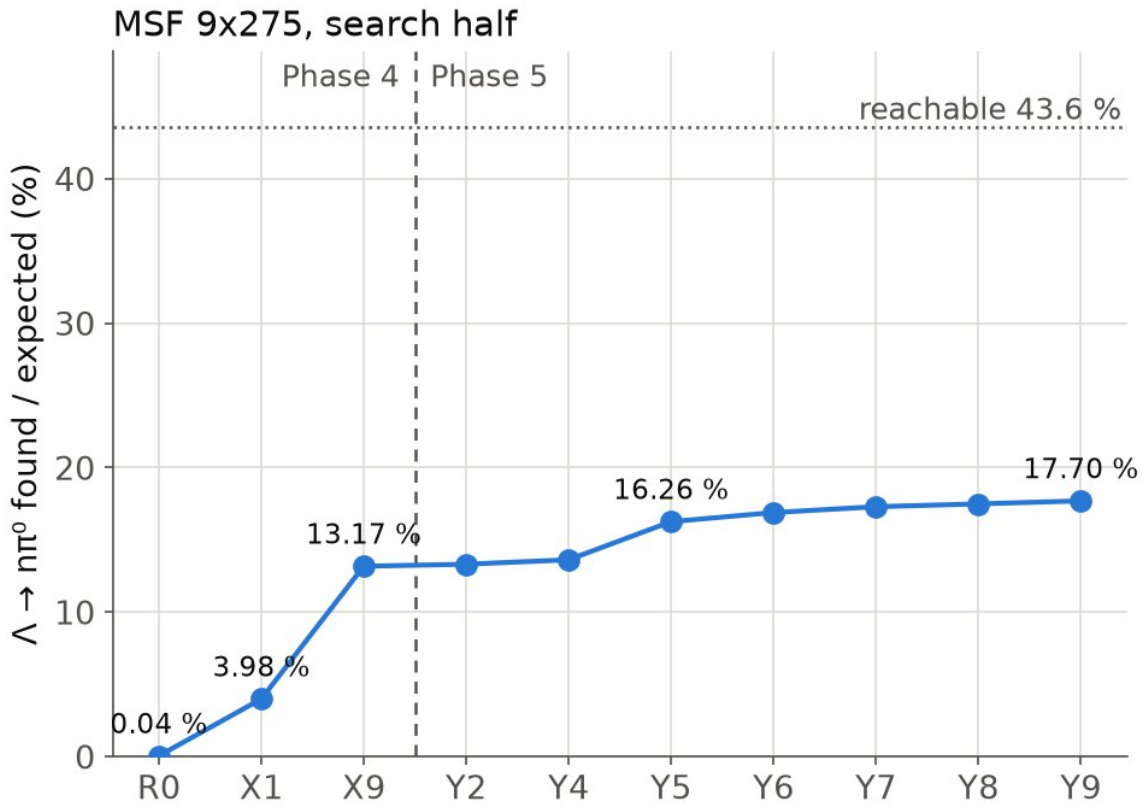
b 28, c 123, z +7.73; purity z_P +2.48

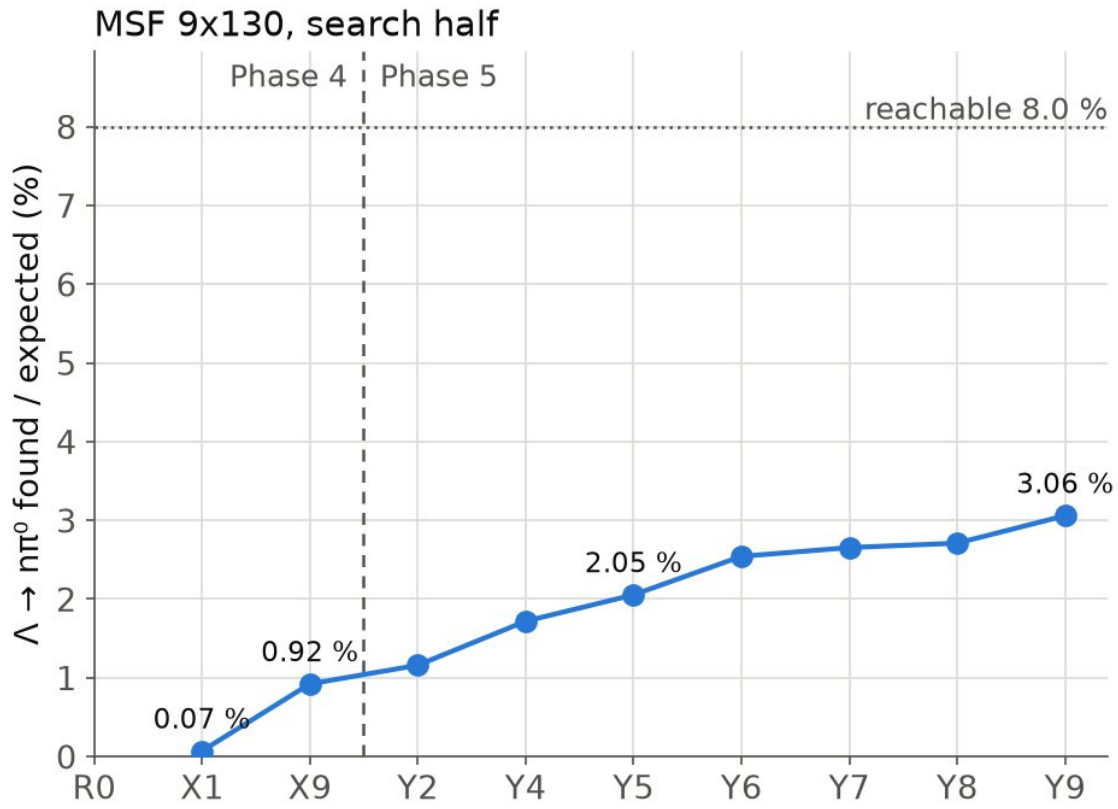
D4 Λ gun, S4a_tm per thrown Λ → nπ⁰ (reported, never selected on)

- X9 (end of Phase 4): 7.44 % mean, 14.10 % at 275 GeV; Y9: 7.44 % mean, 14.12 % at 275 GeV (ceilings 14.1 % / 27.8 %)

Y7–Y9 measured again on files nothing was tuned on.

Phase 4 and Phase 5 on the MSF score — yield step by step

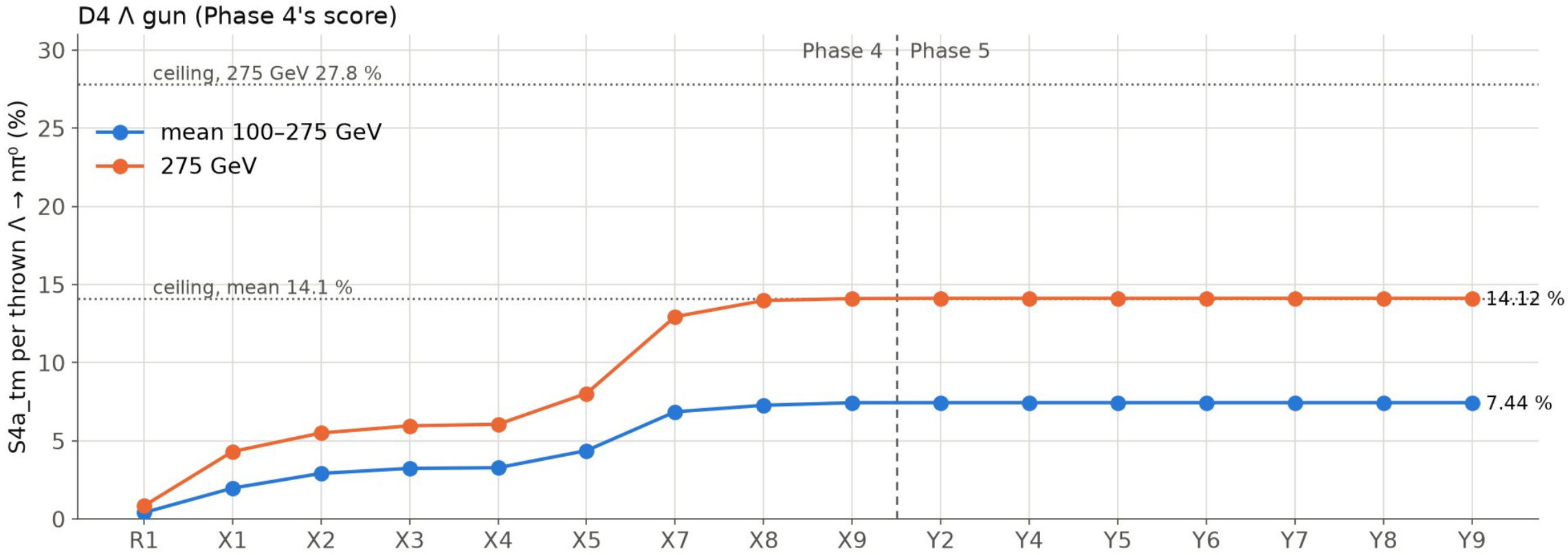




Λ → nπ⁰ found per expected on the MSF search halves (210). R0 the pin; X1, X9 Phase 4 rows (tuned on the D4 gun); Y2–Y9 the accepted Phase 5 tries (tuned on MSF). R0 was not run at 9x130.

Phase 4 took MSF 9x275 from 0.04 % to 13.2 % without seeing MSF; Phase 5 added 4.5 points there and tripled 9x130.

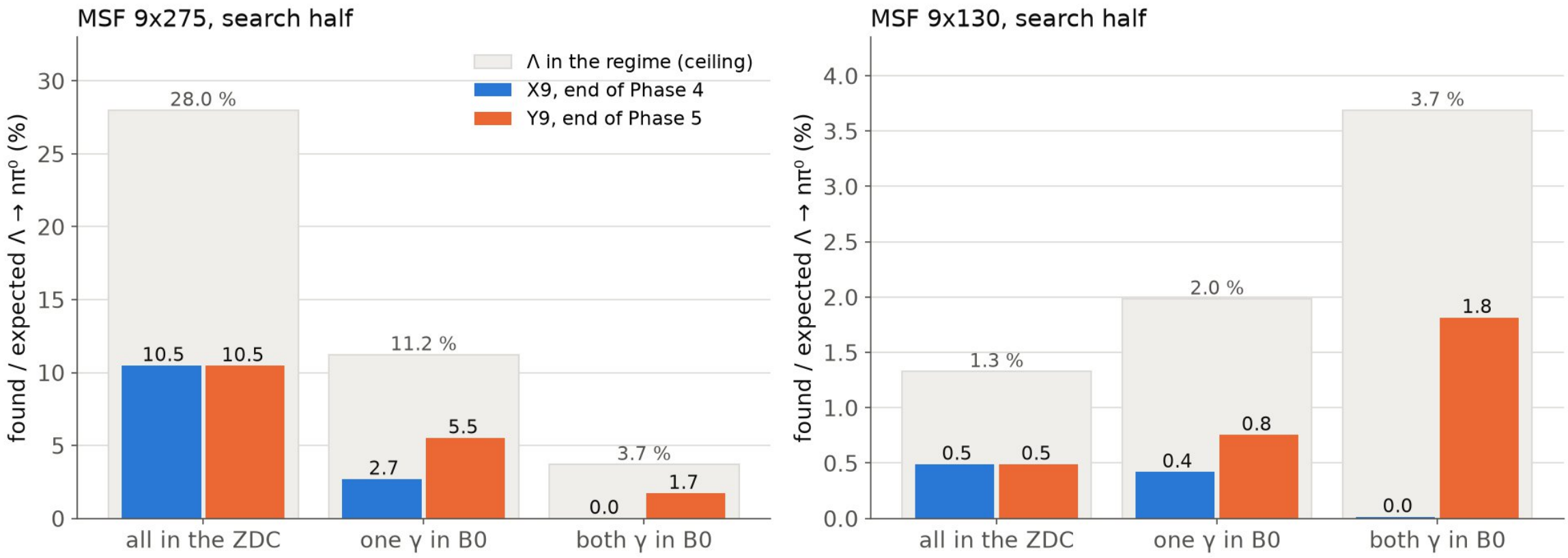
Phase 4 and Phase 5 on the D4 gun score — S4a_tm step by step



S4a_tm per thrown Λ → nπ⁰ on the D4 Λ gun (headline.csv): Phase 4 tries R1 → X9, then every accepted Phase 5 try (D4P5_*, CP5_1_Y6_D4, CP5_2_Y9_D4).

Phase 5 leaves the gun where Phase 4 left it: its gains are in B0 regimes the gun barely reaches.

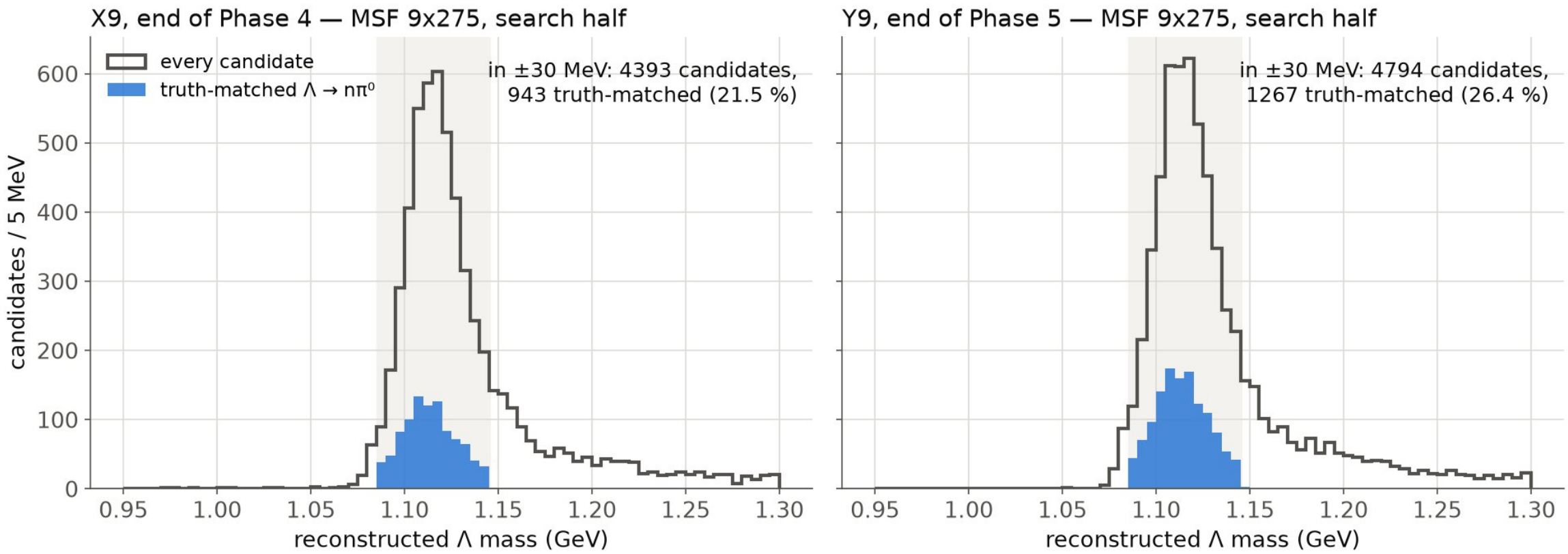
Phase 5's gain by regime — X9 and Y9 on MSF



Found per expected Λ → nπ⁰ in each regime (210 groups, search halves); grey: the regime's share of expected, the most any reconstruction could find there.

All-ZDC unchanged; one photon in B0 doubled; both photons in B0 from zero to 1.7 % (9x275) and 1.8 % (9x130).

Λ mass on MSF 9x275 — every candidate and the truth-matched ones, X9 and Y9



ReconstructedLambdas mass, at most one candidate per event (F3a), MSF 9x275 search half; shaded: the ±30 MeV scoring window; blue: candidates 210 truth-matches to the true n, γ, γ.

The wrong candidates peak at m_Λ too (both pairings rank by Λ mass): purity needs other variables — a phase of its own.

Phase 4 and Phase 5 side by side

	Phase 4 — the Λ bar fight	Phase 5 — Λ → nπ⁰ on MSF events
data it tuned on	D4 Λ gun (8 momenta × 2000)	MSF 2026-11 search halves (2 × 20 000 events)
score	S4a_tm per thrown Λ → nπ⁰ (D4)	N_found on MSF, purity as a floor
held out	D4h (fresh seeds)	D6h (MSF files 41–80)
tries	9: 8 accepted, 1 rejected	9: 7 accepted, 1 inconclusive, 1 rejected
D4 S4a_tm, mean / 275 GeV	0.404 % / 0.832 % → 7.436 % / 14.101 %	X9 → Y9: 7.439 % / 14.117 % (reported, not selected on)
MSF 9x275 found / expected (search)	R0 0.04 % → X1 4.0 % → X9 13.2 %	X9 13.2 % → Y9 17.7 % (held out 12.5 % → 16.8 %)
MSF 9x130 found / expected (search)	X9 0.92 %	X9 0.92 % → Y9 3.06 % (held out 0.88 % → 2.86 %)
purity in the window, 9x275 / 9x130	X9 21 % / 9 %	Y9 26 % / 29 %
what changed	ZDC clustering, photon tag, Λ pairing	vertex lever arm, pairing with B0 photons, B0 splitting and law

Next: Phase 6, the crystal-seeded ZDC photon finder (prepared, loop v6)

- the all-ZDC regime loses a photon for 1125 of 2003 Λ (9x275 half); a scope study decides first (stop below +50 emulated)